



Department of Computer Science and Engineering

Academic Year 2024 – 2025 (Odd Semester)

Degree, Semester & Branch: I Sem – B.E. Computer Science and Engineering - B

Course Code & Title: GE3151 Problem Solving and Python Programming

Name of the Faculty member (s): Dr.R.Venkatesh, ASP/CSE

Innovative Practice Description

- **Unit / Topic: Unit IV / Dictionaries: operations**
- **Course Outcome: CO4**
- **Topic Learning Outcome: TL11**
- **Activity Chosen: One Minute Paper**
- **Justification:**
 - The One Minute Paper activity is ideal for teaching Dictionaries: operations because it encourages students to reflect on their understanding of key operations like insertion, deletion, and iteration.
 - It promotes active learning by summarizing core concepts concisely, helping instructors identify areas where students may need additional support.
 - Additionally, it fosters critical thinking by prompting students to articulate their understanding of dictionary functionality within a limited time.
- **Time Allotted for the Activity: 10 Minutes**
- **Procedure of the Implementation:**
 - Faculty, begin the session by briefly explaining what a dictionary is in programming, its structure, and common operations like adding, updating, deleting, and iterating key-value pairs and provided a simple code example to demonstrate these operations.
 - Introduce the One Minute Paper activity and explain its purpose: to reflect on key dictionary operations and assess understanding.
 - Emphasize the importance of being concise and accurate in their responses.
 - Share a specific question or prompt with the students. Examples include:
 - "Explain how to add a key-value pair to a dictionary with an example."
 - "What happens if you try to access a key that doesn't exist in a dictionary? How can this be handled?"
 - Give students exactly one to five minutes to write their responses individually as shown in fig 1.
 - The faculty member collected and corrected the syntax as shown in fig 2. The students who have written wrong procedure were asked to rewrite it.
 - Ensure the environment is quiet to allow focused thinking.

• CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO5	PO9	PO10	PSO2
CO2	3	2	1	1	1	1	2

(1 – Low 2 – Moderate 3 – High)

• PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1
Justification for correlation	Students will be able to remember the concepts well which is very much needed for a programming.	Students will be able to apply the concepts while approaching the coding practices	Students will be able to provide the solution for the dictionary related problem.	As the student involved in the activity as an individual, the individual work skill improved	Students written/ documentation skill will be improved as they are writing the Dictionaries: operations.	The problem solving skill earned through this activity helps the students in solving various problems in information and communication engineering.

• Images / Screenshot of the practice:



Fig 1: One-Minute Paper Activity

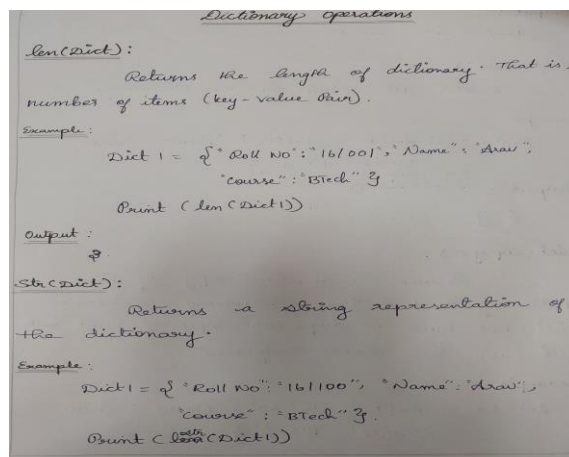


Fig 2: Sample Sheet of activity of the Student Kamalee J J

• Reflective Critique:

❖ Feedback of practice from students and other stakeholders:

- Students often appreciate the concise nature of the activity, as it allows them to quickly assess and articulate their understanding.

- They value the immediate clarification of doubts and enjoy the interactive discussion of their responses.

❖ ***Benefit of the practice:***

- Students can articulate and reflect on their understanding of dictionary operations, helping them solidify concepts like adding, updating, deleting, and accessing data in dictionaries.
- Instructors can quickly ensure student understanding and address misconceptions in real time, ensuring effective learning.
- The short timeframe encourages students to focus on key concepts and think critically about how dictionary operations work.

❖ ***Challenges faced in implementation:***

- Some students might struggle to condense their understanding into a concise response, particularly if they are less confident in the topic.
- Advanced learners might find the prompts too simplistic and require more challenging questions.
- Reviewing and analyzing all responses can be overwhelming in larger classes.
- The short timeframe may lead to surface-level responses that don't fully reflect students' understanding or misconceptions.

References:

- <https://www.slideshare.net/raja2dlas/minute-paper>
- <https://www.ritrjpm.ac.in/images/computer-science/CS8251-SE-MinutePaper.pdf>

Signature of Faculty Member

HOD-CSE